

Best practices for moving from correlation to causation in ecological research

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In ecology, causal questions are ubiquitous, yet the literature describing systematic approaches to answering these questions is vast and fragmented across different traditions (e.g., randomization, structural equation modeling, convergent cross mapping). In our Perspective, we connect the causal assumptions, tasks, frameworks, and methods across these traditions, thereby providing a synthesis of the concepts and methodological advances for detecting and quantifying causal relationships in ecological systems. Through a newly developed workflow, we emphasize how ecologists' choices among empirical approaches are guided by the pre-existing knowledge that ecologists have and the causal assumptions that ecologists are willing to make.

Causality in ecological studies

Ecology is centered around investigating causal relationships between living organisms and their environments. In ecology, as in many other scientific fields, causality is understood as a phenomenon where a change in one variable (the cause) induces a change (the effect) in another variable^{1–4}. Thus, a causal relationship between X and Y exists if a perturbation in the cause X produces a change in the responding variable Y ^{4,5}, potentially through the perturbations of intermediary variables^{6,7}. This perturbation-based definition of causality is the definition most familiar to scientists and philosophers^{4,8}.

Because of a strong tradition of using manipulative experiments to establish causation, ecology has been shaped by two aphorisms: correlation does not equal causation, and causal claims can only be

made from experiments. The first aphorism oversimplifies the complexity of causal relationships and has been critiqued in the literature^{5,9,10} – correlation does not always equal causation, but correlation can suggest a causal relationship (see the “Using assumptions to move from correlation to causation” section). More importantly, the first aphorism does not imply the second: imperfectly designed experimental studies can mistakenly suggest causal relationships where none exist, and causation can, in fact, be established through well-designed observational studies^{11–13}. Natural history approaches, for instance, have long been used to establish credible causal claims (e.g., sea otters driving trophic cascades in subtidal communities^{14,15}). Recently, interest in observational approaches has grown^{16,17} due to the economic, ethical, and logistical challenges of manipulating ecological

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variables¹⁸ and the limitations of experiments in capturing complex, large-scale causal relationships in nature¹⁹. Observational data, particularly from multiple locations and time points, are increasingly valued for complementing experiments and supporting more generalizable causal claims^{19–21}.

To formalize the requirements for making causal claims from experimental and observational data, scholars in various fields have made substantial advances in mathematical and statistical tools over the past 50 years^{12,22–28}. Applications of these advances have changed how we think about scientific topics such as environmental and genetic causes of disease^{29–31}, military veterans' health³², criminology^{33,34}, and education^{35,36}, and have influenced policies on air pollution^{37,38} and carcinogens³⁹. These same advances are increasingly being proposed by ecologists to investigate causal questions using observational^{9,27,40–49} and experimental data^{50–52}.

Yet the way in which these advances relate to each other is not readily apparent from the published literature. For example, what are the conceptual connections between studies that use experimental designs and studies that use convergent cross-mapping algorithms? Published reviews typically focus on one set of approaches at a time (e.g., quasi-experimental designs, structural causal models, dynamical systems)^{27,41,44,53,54}, which makes it difficult for ecologists to understand how, or if, the seemingly disparate approaches are related.

In this Perspective, we connect the assumptions, tasks, frameworks, and methods across these approaches, thereby providing a synthesis of the concepts and methodological advances for detecting and quantifying causal relationships in ecological systems. When answering a causal question, we must first identify the appropriate causal task: either causal discovery, which focuses on detecting whether causal relationships are likely to exist between variables in a system, or causal inference, which focuses on quantifying the direction and magnitude of causal relationships without bias. To accomplish these tasks, we employ causal frameworks, such as the structural causal model framework¹², the potential outcomes framework²⁵, or the dynamical systems causality framework^{55,56}, which formally define causal relationships and specify the assumptions that must be satisfied to accurately detect or quantify causal relationships from data. These frameworks then guide the selection of causal methods, that is, study designs and algorithms, which are used to operationalize these assumptions and establish the conditions necessary to make causal claims.

To outline the process of navigating tasks, frameworks, and methods, we created a workflow of best practices for answering causal questions in ecological research. To provide further readings and software to implement the ideas in the Perspective, we provide comprehensive Supplementary Information.

Throughout our Perspective, we highlight how well-articulated causal assumptions are the glue that unifies the myriad approaches to answering causal questions in ecology. These assumptions, together with the research question, shape every decision in the workflow – guiding which pre-existing knowledge is relevant, which causal task is

pursued, and which study design or algorithm is implemented. Because understanding these assumptions is a prerequisite for using the workflow, we provide a clear articulation of the fundamental causal assumptions required to move from correlation to causation. These assumptions also facilitate transparent discussions about the adequacy of the study designs and algorithms that help scholars move from observations of statistical dependence in data to claims about causal relationships in ecological systems.

Using assumptions to move from correlation to causation

Data never speak by themselves. To derive meaningful causal insights from data, we must rely on well-defined hypotheses, statistical models grounded in ecological theory, and both testable and untestable assumptions^{57–59}. The importance of hypotheses, appropriate statistical models, and statistical assumptions is well known in ecology.

Less well known is the importance of causal assumptions that allow researchers to go from making claims about correlations to making claims about causation. Unlike most statistical assumptions, causal assumptions are typically untestable^{60–62}; that is, causal assumptions cannot be verified from data, even unlimited data^{57,58}. For example, experimentalists assume that randomization of a treatment ensures that any differences in outcomes across the randomized groups can only be attributed to the treatment or sampling variability⁵⁰. Yet, this assumption cannot be verified. Causal assumptions, when combined with principles of probability theory and statistical dependence, allow us to make causal claims from data. The formalization of these assumptions is one of the most important scientific advances for answering causal questions^{26,28,58}. For more details on the contrast between statistical and causal assumptions, see Supplementary Note 1: Causal Versus Statistical Assumptions.

Causal assumptions, in tandem with statistical assumptions about the data structure, establish when statistical dependence can be interpreted as evidence for the perturbation-based notion of causality^{12,25,27,63}. Consider a scenario in which we seek to determine whether, or by how much, variation in the abundance of aphid predators (e.g., ladybird beetles) (X) changes aphid abundance (Y). If our knowledge about the probability of aphid abundance changes after learning something about ladybird beetle abundance, then the two variables are statistically dependent. This dependence forms the starting point for using data to investigate potential causal relationships between two variables.

Statistical dependence is linked to causality through the Common Cause Principle⁶⁴, which states that if a statistical dependence exists between two variables X and Y , then at least one of the following is true: X causes Y , Y causes X , or X and Y are both caused by a third variable C (Fig. 1). In ecology, a commonly used measure of statistical dependence is correlation, which describes the linear similarity between two sets of observations. The presence of correlation can therefore signal a causal relationship. The lack of correlation, however, does not necessarily rule out statistical dependence or causality, as correlation is just one possible measure of dependence between two variables.

In causal analyses, we wish to distinguish variables with direct causal links from those that are not causally influencing each other. Thus, eliminating the possibility that a third variable C causes both X and Y is often a priority (i.e., we seek to eliminate non-causal, rival explanations for statistical dependencies). For example, in Fig. 2, broad-spectrum pesticide use affects ladybird beetle abundance and earthworm abundance. However, beetle abundance does not influence earthworm abundance, nor vice versa. In this case, any observed statistical dependence between beetle abundance (X) and earthworm abundance (Y) is entirely attributable to their common cause, pesticide use (C).

To eliminate these common causes (a.k.a., confounding variables or confounders), researchers make three assumptions: the Causal

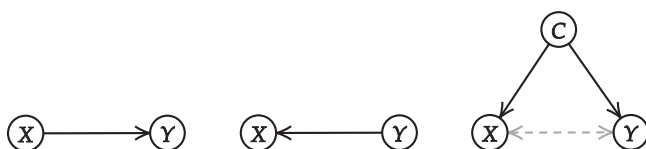


Fig. 1 | Statistical dependence implies three possible causal relationships: X causes Y (left panel), Y causes X (middle panel), or X and Y are caused by a common variable C (right panel). All three relationships can exist simultaneously in many contexts (indicated by the dashed gray arrows). Causal assumptions aim to eliminate the third possibility (right panel) because the presence of C introduces additional statistical dependence between X and Y that is not due to any direct causal relationship.

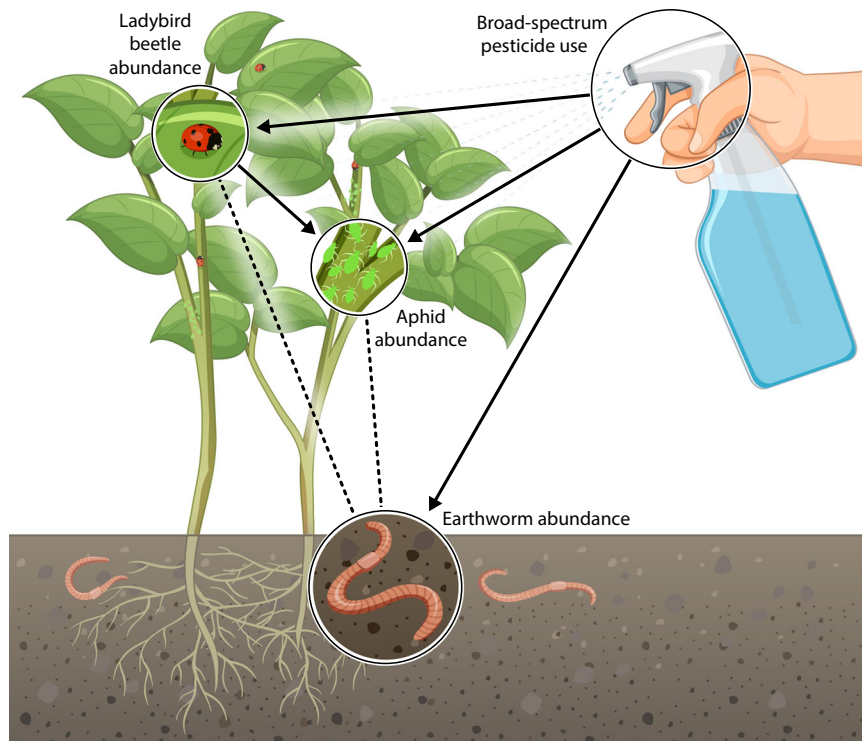


Fig. 2 | Illustration of the common cause principle in an ecological system. In this system, abundance of ladybird beetles, aphids, and earthworms is statistically dependent but not necessarily causally related. Solid arrows represent directional

causal relationships, and dashed lines represent statistical dependences that do not reflect direct causal relationships.

BOX 1

Three fundamental causal assumptions

For these assumptions, we define two variables X and Y as statistically dependent if the probability that Y takes a specific value given that X has taken a specific value is different from the probability that Y takes a specific value without any information about the value that X has taken (i.e., $P(Y|X) \neq P(Y)$). In other words, if X and Y are statistically dependent, knowing something about X changes what is known about the probability of Y .

- Causal Sufficiency⁵² (a.k.a., the no unmeasured confounding assumption^{55–57}) requires that we observe all variables in a set $\mathbf{C}$ that causally influence any pairs of variables X and Y , and we include $\mathbf{C}$ in our model that describes the relationship between X and Y , thus ensuring that no confounding variables are unobserved.
- The Causal Markov Condition^{54,58,59} states that if a pair of variables X and Y are statistically dependent solely because both are caused by a common variable C , and if we control for C by including it in our model, then X and Y become conditionally independent given C .
- Causal Faithfulness⁵², stated very loosely, declares that statistical independence (conditional or unconditional) between a pair of

variables X and Y indicates the absence of a causal relationship between those variables.

The combination of the Causal Markov Assumption (A2) and the Causal Faithfulness Assumption (A3) allows us to claim that if two variables, X and Y , are conditionally independent when C is included in the model, then X and Y are not causally related but instead are caused by a third common variable C . The Causal Sufficiency Assumption (A1) then ensures that we can distinguish causal relationships from dependence induced by a common cause if we include all possible confounders between variables in a model that describes the relationship between X and Y .

The Causal Markov and Causal Faithfulness assumptions have formal definitions requiring technical notation that are beyond the scope of this article. For a full discussion of these assumptions, we refer the reader to Pearl²³ and Spirtes and Zhang⁶⁰.

Sufficiency Assumption²⁸, the Causal Markov Condition^{64–66}, and the Causal Faithfulness Assumption²⁸ (Box 1). Together, these untestable assumptions allow us to distinguish direct causal relationships between variables from dependence between variables induced by a common cause. By including all common causes in a model of the relationship between X and Y (A1 in Box 1), we can eliminate the portion of dependence due to those common causes $\mathbf{C}$ (A2). Any remaining statistical independencies can be interpreted as evidence of

no causal relationship between the variables (A3), while any remaining dependence implies the possibility of a direct causal relationship.

For example, if pesticide use is a common cause of both ladybird beetle abundance and aphid abundance, then we should include pesticide use in a model of the relationship between ladybird beetle abundance and aphid abundance (Fig. 2). If pesticide use is the only common cause and, after conditioning on it, beetle abundance is statistically independent of aphid abundance (i.e., they are conditionally

independent), then, under the three causal assumptions, we can infer that no causal relationship exists between them. Conversely, if beetle abundance and aphid abundance are not independent conditional on pesticide use, then a causal relationship between beetle abundance and aphid abundance may exist (i.e., a lack of conditional independence means we cannot rule out a causal relationship, but it does not provide definitive evidence of causation).

The three causal assumptions required to connect statistical dependence to causal dependence – Causal Sufficiency, Causal Markov Condition, and Causal Faithfulness – are the foundation upon which causal claims are made from experimental and observational data. These causal assumptions allow us to differentiate the causal dependencies between two variables from the non-causal dependencies created by confounding variables.

Satisfying causal assumptions with pre-existing knowledge, study designs, and algorithms

Given the restrictive and untestable nature of the three causal assumptions introduced in the previous section, ecologists may wonder whether causal claims can realistically be made from ecological data, since satisfying these assumptions requires building models that account for all confounders. Unlike models built for prediction or description, models built to make causal claims cannot be validated using goodness-of-fit or predictive accuracy metrics, as these metrics assess how well a model describes the observed data but do not evaluate how well the model satisfies the untestable assumptions required for making causal claims^{67,68} (for more details, see Supplementary Note 2: Defining Causal and Non-Causal Research Questions). In the following subsections, we describe how the foundations for satisfying causal assumptions are provided by pre-existing knowledge, study designs, and algorithms.

Pre-existing knowledge

To determine if the three causal assumptions can be satisfied in a specific study context, pre-existing knowledge is essential⁵⁹. Pre-existing knowledge guides us in our efforts to identify potential confounders and other potential factors that produce variation in the causal variable(s) (see the “Summarize pre-existing knowledge” section). It helps us determine which confounders can be measured^{43,69,70} and which are likely unobservable, a determination that guides the choice of study designs or algorithms (see the “Study designs and algorithms” section). Pre-existing knowledge can also be used to detect or rule out the presence of uncontrolled confounders through, for example, falsification tests (see the “Select a study design or algorithm, apply estimation methods, obtain results, and interpret results” section). The more pre-existing knowledge to which we have access, the stronger the causal claims that we can make from an analysis of data (i.e., the more plausibly we can eliminate non-causal, rival explanations for statistical dependencies).

Pre-existing knowledge can include general and domain-specific ecological theory, subject matter expertise, field experience, and findings from other studies, including studies that use empirical approaches lacking causal interpretations (see Supplementary Note 2: Defining Causal and Non-Causal Research Questions). Because pre-existing knowledge is often complex and wide-ranging, we need succinct and straightforward ways to summarize it. In the “Summarize pre-existing knowledge” section, we describe two common tools for organizing our understanding of an ecological system (i.e., our “mechanistic knowledge”⁷⁰).

Study designs and algorithms

Pre-existing knowledge is typically not sufficient to satisfy causal assumptions. For instance, even if we can identify all confounders with pre-existing knowledge, we are unlikely to be able to measure them all, which would be necessary to satisfy the Causal Sufficiency

Assumption. However, study designs and algorithms provide us with the opportunity to address such challenges by relaxing one or more of the three causal assumptions in the “Using assumptions to move from correlation to causation” section in favor of equally untestable but (hopefully) more plausible causal assumptions.

Experimental designs, for example, substitute the Causal Sufficiency Assumption with the assumption that treatment randomization eliminates the effects of unmeasured confounding variables^{25,71}. Confounders are thus addressed through design rather than measurement. In non-experimental studies, observational designs often relax the Causal Sufficiency Assumption through statistical techniques that define the minimum set of confounding variables that need to be observed to accomplish the desired causal task^{57,58,72}, or through statistical techniques that allow researchers to pursue alternative research goals that reduce the number of confounders that must be measured (e.g., by defining alternative causal effects^{61,73} or by detecting and accounting for possible unmeasured confounders⁵³). These statistical techniques and redefined research goals can also be used with experimental designs that face implementation challenges, such as when the experimental manipulation affects the outcome variable through other pathways (i.e., randomization is a confounder), or when post-randomization observations are missing (i.e., attrition). We provide more details on both experimental and observational study designs and algorithms in the “Select a study design or algorithm, apply estimation methods, obtain results, and interpret results” section.

A workflow for answering causal questions in ecology

We present a comprehensive workflow that summarizes the best practices for systematically addressing causal questions in ecology (Fig. 3). The workflow serves as a roadmap, beginning with the causal question and ending with the interpretation and validation of results. Each step in the workflow marks a decision point that reflects a best-practice principle that ensures our causal research is robust, transparent, and aligned with the assumptions necessary to make causal claims from statistical analyses of ecological data. The workflow is designed to be flexible, so ecologists can adapt it to their pre-existing knowledge, data, and preferred methods.

Here, we summarize the workflow steps, and we elaborate on them in the subsequent four sections:

- 1. Define the Causal Question and Summarize Pre-Existing Knowledge:** *Before any data are collected or examined, develop a clear, testable causal hypothesis and describe all potential confounders.* We first define the causal research question with at least one outcome variable (Y) and one or more hypothesized causal variables (X) (see Supplementary Note 2: Defining Causal and Non-Causal Research Questions for differences between causal and non-causal questions in ecology). Then, to identify all confounding variables, we assess the corpus of pre-existing knowledge on the causes and outcomes of interest. We can summarize this knowledge using causal diagrams or thought experiments.
- 2. Define the Causal Task:** *Choose a causal task that matches the study’s causal question and the depth of available prior knowledge.* When answering causal questions, we use pre-existing knowledge to determine whether to pursue causal discovery or causal inference. Causal inference, which seeks to quantify the magnitudes of causal relationships, is feasible when we have sufficient pre-existing knowledge to be confident of the causal, outcome, and confounding variables and the directions of the causal relationships. If this knowledge is insufficient, we can instead pursue causal discovery, which aims to detect the existence and direction of causal relationships.

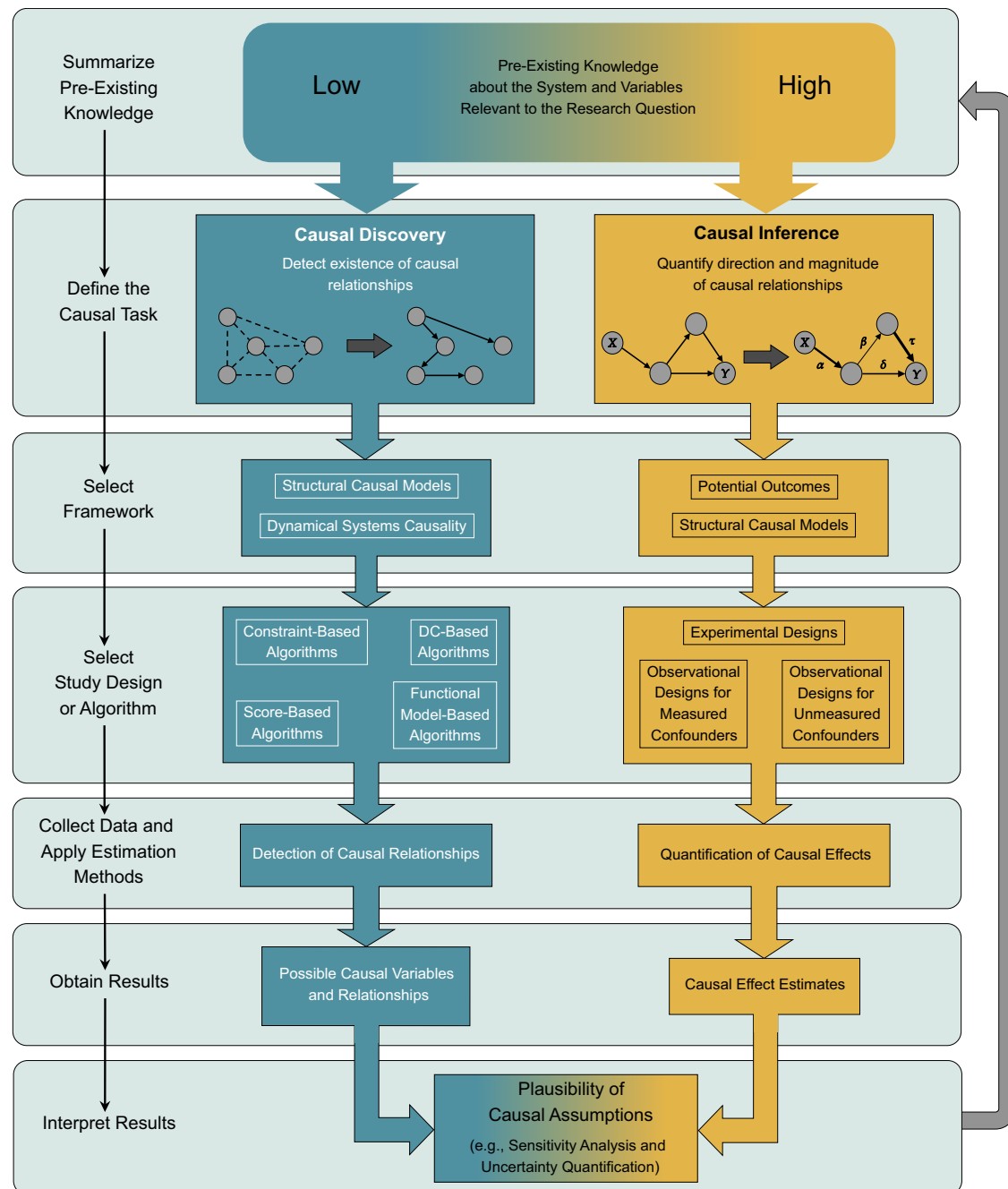


Fig. 3 | A best-practice workflow that outlines the key steps and decisions for answering causal questions in ecological research. The workflow illustrates a step-by-step process for conducting causal analyses. Arrows indicate the typical flow of an analysis. Two possible pathways are shown: causal discovery approaches (blue), which aim to identify the existence of causal relationships when pre-existing

knowledge is low, and causal inference approaches (yellow), which aim to quantify the direction and magnitude of causal effects when pre-existing knowledge is high. The gray feedback loop on the right highlights the iterative refinement of causal analyses based on assessments of the plausibility of causal assumptions.

3. Select Framework: Adopt a formal causal framework through which causal assumptions can be explicitly articulated, tested, and communicated. To clearly articulate the causal and statistical assumptions that must be satisfied for valid claims in either causal task, we can use one or more causal frameworks. The potential outcomes framework and the structural causal model framework are two common frameworks used for causal inference. For causal discovery, the structural causal model and dynamical systems causality frameworks are frequently used.

4. Select Study Design or Algorithm, Collect Data and Apply Estimation Methods, Obtain Results, and Interpret Results: Select a study design or algorithm that aligns with the study's articulated causal assumptions, and then rigorously assess the plausibility of those assumptions and implications of violations of those assumptions when drawing conclusions. For causal inference, study designs can be grouped into three categories: experimental designs, observational designs for measured confounders, and observational designs for unmeasured

BOX 2**Ecologists conducting causal research using the best-practice workflow in Fig. 3****Define the Causal Question and Summarize Pre-Existing Knowledge**

An intertidal ecologist seeks to quantify the change in bivalve abundance (Y) caused by floods (X) through changes in nitrogen (M_1) and salinity (M_2) in intertidal zones at the mouth of an estuary. The ecologist summarizes knowledge about all confounders for each of the causal relationships of interest (i.e., floods on bivalves, floods on nitrogen, floods on salinity, nitrogen on bivalves, and salinity on bivalves).

A tiger ecologist seeks to determine the ecological factors (X) that encourage tigers to make more visits or spend more time (Y) in certain locations. The ecologist summarizes knowledge about confounders of the causal relationship between ecological factors and tiger occupancy (e.g., geographic and human factors).

Define the Causal Task

The intertidal ecologist has robust ecological theory and a large collection of prior studies to identify the full set of confounders that could bias estimation of any one of the causal relationships of interest. Thus, the ecologist decides to pursue causal inference.

The tiger ecologist has theory and field observations to identify some ecological factors that may influence tiger occupancy, but they do not have sufficient knowledge to identify all human and geographic confounders. Thus, the ecologist decides to pursue causal discovery.

Select Framework

The intertidal ecologist adopts the structural causal model framework, which they prefer for its structural approach to reasoning about multiple causes jointly.

The tiger ecologist adopts the dynamical systems causality (DC) framework to accommodate the complex and evolving dynamics of their study system.

Select Study Design or Algorithm

The intertidal ecologist selects an observational study design in which they measure and condition on all confounders.

The tiger ecologist selects a DC-based algorithm appropriate for causal discovery when many confounders are unmeasured.

Collect Data and Apply Estimation Methods

The intertidal ecologist collects observational cross-sectional data on all causal, outcome, and confounding variables related to the causal relationships of interest and then fits a structural equation model to quantify the causal relationships of interest.

The tiger ecologist collects observational time-series data for tiger occurrence, abundance of prey species, poaching activity, and weather conditions at a series of locations and uses convergent cross-mapping (CCM) to detect causal relationships between pairs of variables.

Obtain and Interpret Results

The intertidal ecologist obtains estimates of the causal effects of floods on bivalve abundance that arise through the changes in nitrogen and salinity. They perform a causal sensitivity analysis that quantifies how much the estimates change in the presence of an unmeasured confounder.

The tiger ecologist obtains a network with detected causal relationships between pairs of variables. They perform a sensitivity analysis that shows how the detected causal relationships change when the CCM hyperparameter settings are changed.

confounders. Within these categories, many approaches exist (e.g., regression adjustment⁷⁴, propensity score matching^{45,75}, and structural equation modeling⁹). For causal discovery, algorithms are used instead of study designs. These fall into four categories: constraint-based, score-based, functional model-based, and dynamical systems causality-based. Within these categories, many algorithms are available (e.g., convergent cross-mapping²⁷, fast causal inference²⁸, and greedy equivalency search⁷⁶). Based on the requirements of the study design or algorithm, we then collect data and apply estimation methods to detect causal relationships or quantify causal effect(s). Afterwards, we interrogate the plausibility of the causal and statistical assumptions by identifying potential violations of the assumptions and exploring the implications of those violations for the conclusions.

To illustrate the workflow's application to real-world ecological research, we use two example ecologists, an intertidal ecologist and a tiger ecologist. In Box 2, we summarize how each ecologist navigates the workflow.

Although we present the workflow in a linear fashion, researchers can use it iteratively in two ways: (i) the results from one causal analysis will feed into future analyses in the form of pre-existing knowledge⁷⁰ (gray arrow in Fig. 3); and (ii) after taking actions at one step, researchers may need to return to previous steps before advancing in the workflow (e.g., refining the causal question if prior knowledge is insufficient; reassessing the study design if data collection did not go as planned).

Summarize pre-existing knowledge

One common conceptual tool for summarizing pre-existing knowledge is a causal diagram. Causal diagrams help us organize our pre-existing knowledge by visually mapping the presumed causal relationships among causes (X), their outcomes (Y), and confounders (C). The most widely used causal diagram is the causal directed acyclic graph (causal DAG), which follows a set of formal rules that define how causal relationships must be encoded⁷⁷. A causal DAG includes the focal variables of a study (i.e., the cause and the outcome variables), along with all suspected common causes (i.e., confounders) between the focal variables. Directed edges (arrows)

between variables indicate that unidirectional causal relationships are presumed to exist, and the absence of an arrow between two variables reflects a strong assumption that a causal relationship does not exist¹². Causal DAGs, which must include all potential confounders of presumed causal relationships, enable us to identify the confounders we need to address with experimental or statistical techniques. Thus, causal DAGs should be constructed at the beginning of a study, before data are collected and the specific study design or algorithm is chosen.

Some ecologists will be familiar with the structural equation model (SEM) diagram⁹, which can be interpreted as a causal DAG when its structure represents only unidirectional relationships and explicitly encodes assumptions about causal relationships, including all relevant confounders^{78,79}. SEM diagrams also include additional parametric assumptions and are purpose-built for SEM analyses⁷⁸, whereas causal DAGs, which require no assumptions about the functional forms of causal relationships, can be used in any type of causal analysis.

Another conceptual tool for summarizing pre-existing knowledge is a thought experiment in which researchers consider how they would conduct a hypothetical ideal randomized controlled trial (RCT) – often termed a target trial^{80,81} – to answer their causal research question²⁵. By comparing the ideal (target) trial with the actual data-generating process, we can identify discrepancies that may lead to bias through confounding variables that distort the observed relationship between the causal variable and the outcome. Formulating such a target trial forces us to articulate all the key components of an ideal RCT and then systematically determine which of these components may be absent or imperfect in our study. In doing so, it becomes clearer which variables, including potential confounders, should be accounted for in the analysis to emulate the conditions of an ideal experiment. Just as drawing causal DAGs helps visualize the network of causal relationships and identify confounders, formulating these thought experiments provides a concrete tool for planning rigorous study designs (i.e., the thought experiment forces us to ask the question, “From where does the variation in the causal variable come?” or, equivalently, “What is the treatment assignment mechanism?”). For resources that describe how to draw causal DAGs or develop RCT thought experiments for studies, see Supplementary Note 3: Formally Summarizing Pre-Existing Knowledge.

Define the causal task – causal discovery or causal inference

In choosing the most appropriate causal task for a research question, we must carefully consider the gap between available knowledge and the knowledge that would be required to plausibly satisfy causal assumptions. When pre-existing knowledge is extensive, we may pursue the task of causal inference. When pre-existing knowledge is limited, we may instead pursue causal discovery. Although the dividing lines between these two tasks are not as clear-cut as implied in our workflow (i.e., causal research lies on a continuum rather than in one of two camps), the contrast between their goals is illuminating for understanding how each task draws on pre-existing knowledge.

The goal of causal inference is to quantify the magnitudes of causal effects, such as the effect of a change in temperature of one degree on wildlife mortality or the effect of the introduction of a wildfire suppression program on tree species composition. Causal inference requires substantial pre-existing knowledge about which variables act as causes, outcomes, and confounders, as well as the directions of causal processes (high pre-existing knowledge in Fig. 3). Quantifying multiple causal effects within an ecological system is even more challenging because sufficient pre-existing knowledge must exist to satisfy the required causal assumptions for every pair of cause–outcome variables.

To quantify causal effects, all causal strategies begin by defining the specific effect(s) of interest that connect theoretical quantities to

data. Different causal effects require different variations of the causal assumptions⁸². Ecologists are often interested in the average effect of X on Y across all observations, that is, the average change in the outcome Y per unit change in X . However, other effects may also be relevant, such as average effects for subgroups⁸³ and mediation effects⁸⁴ (effects of intermediary variables between a cause and its outcome). Moreover, some causal effects may be preferred because the causal assumptions for these effects can be more plausibly satisfied for a study (e.g., complier average causal effects, local average treatment effects, etc.).

In contrast to causal inference, causal discovery aims to detect or learn causal relationships among measured variables, such as whether there is a causal relationship between sardine and anchovy populations and in what direction(s). Although causal discovery requires causal assumptions, the assumptions are less restrictive than they are in causal inference, and thus, less pre-existing knowledge is required (low pre-existing knowledge in Fig. 3). While causal discovery methods offer flexibility in investigating causal questions with limited pre-existing knowledge, this advantage comes with the trade-off of potentially less precise or less certain conclusions about causal relationships. Causal discovery is therefore primarily valuable for generating more knowledge to guide subsequent studies.

To detect causal relationships, all causal discovery strategies begin by defining an initial causal diagram (see the “Summarize pre-existing knowledge” section) and then refining it with statistical evidence from data. One strategy begins with a causal diagram that assumes causal relationships exist among all variables. Statistical independence tests are then systematically applied to eliminate connections between variables where evidence of a causal relationship is not supported by the data³³. Another strategy starts with a causal diagram that assumes no causal connections among variables and iteratively adds them where statistical evidence suggests a potential causal relationship⁸⁵. Both strategies rely on variations of the three causal assumptions introduced in the “Using assumptions to move from correlation to causation” section and aim to produce a refined causal diagram that reflects only the causal relationships consistent with the observed data and the underlying assumptions.

Select a causal framework

Both causal inference and causal discovery rely on untestable causal assumptions (see the “Using assumptions to move from correlation to causation” section) that allow researchers to interpret statistical patterns as evidence of causation. Causal frameworks structure how these causal assumptions are represented for a given task, ensuring consistency among study design/algorithm, data collection, and estimation procedures.

Causal frameworks for causal inference

For causal inference, assumptions and estimation procedures are expressed using one of three causal frameworks: the Neyman–Rubin causal model, also commonly known as the potential outcomes (PO) framework; the structural causal model (SCM) framework; and the decision-theoretic framework²². We focus on the PO and SCM frameworks, but readers interested in the decision-theoretic framework can refer to Dawid²² and Dawid⁸⁶.

The choice of framework is primarily based on researcher preferences, as the PO and SCM frameworks have been shown to be logically and mathematically equivalent^{87–89}. The PO framework may appeal to experimentalists because it expresses causal assumptions by approximating the conditions that most accurately represent an idealized randomized controlled experiment. Alternatively, researchers who primarily model ecological systems as collections of simultaneously interacting variables may prefer the SCM framework, which represents systems as causal DAGs. Structural equation modeling,

when used to make causal claims under causal assumptions^{9,46}, is a subset of the SCM framework^{79,90}.

The ways in which the PO and SCM frameworks express causal assumptions for causal inference are described in Supplementary Note 4: Causal Assumptions Translated through Causal Frameworks and Box S1. Resources for learning more about the core concepts of the PO and SCM frameworks can be found in Supplementary Note 5: Core Concepts for Each Causal Framework.

Causal frameworks for causal discovery

For causal discovery, the assumptions and estimation procedures are expressed using either the SCM framework or the dynamical systems causality (DC) framework^{55,56}. Causal discovery using the SCM framework is well-suited for ecological systems with multiple interacting variables, where causal relationships are expected to be stable across observations. SCM-based causal discovery algorithms also allow researchers to incorporate pre-existing knowledge by specifying constraints on potential causal relationships, making them particularly useful for exploratory studies where some causal relationships are known or hypothesized. In contrast, the DC framework may be more suitable for complex dynamic systems where causal effects unfold over time and cannot be represented as static combinations of causes. DC-based algorithms typically use time-series data to infer causal relationships by testing whether knowledge of one variable's past improves the ability to anticipate changes in another variable. Measures of improvement are typically comprised of changes in predictability or statistical dependence, including those estimated by information-theoretic measures^{85,91}. In both the SCM and DC frameworks, multiple causal diagrams can be consistent with the same structure of statistical dependencies in data, but pre-existing knowledge can refine the causal diagrams by constraining what relationships are possible.

The ways in which the SCM and DC frameworks express causal assumptions for causal discovery are described in Supplementary Note 4: Causal Assumptions Translated through Causal Frameworks and Box S2. Resources for learning more about the core concepts of the SCM and DC frameworks can be found in Supplementary Note 5: Core Concepts for Each Causal Framework.

Select a study design or algorithm, apply estimation methods, obtain results, and interpret results

Study designs for causal inference and algorithms for causal discovery provide structured approaches for satisfying or relaxing the untestable causal assumptions through decisions about the data and analysis (i.e., designs and algorithms operationalize causal frameworks). Designs and algorithms also lead us to appropriate methods for estimation and interpretation of the results.

This section provides an overview of key study designs and algorithms. Their details and applications are beyond the scope of this Perspective, but in Supplementary Note 6: Study Designs and Algorithms for Causal Analyses, we provide resources, including guidance on implementation and relevant software packages. While we focus on foundational study designs and algorithms, we summarize in Supplementary Note 7: Advanced Methods for Causal Inference and Causal Discovery some advanced approaches, including those that integrate machine learning techniques, which are rapidly emerging and may offer new opportunities for causal research in ecology.

Study designs for causal inference

Study designs for causal inference fall into three categories: (1) experimental designs that aim to minimize confounding from both measured and unmeasured variables through manipulation of the causal variable, (2) observational designs that explicitly identify and control for measured confounders, and (3) observational designs that eliminate unmeasured and potentially unknown confounding by leveraging external sources of variation. Representative approaches from these

three categories are listed in Table 1 (references, applications, and available software implementations are provided in Table S6).

Experimental designs are often well-suited for causal inference because they provide a structured approach for directly manipulating the causal variable and defining the temporal order of cause and effect^{50,92}. Through strategies like randomization, we aim to control or eliminate the effects of confounding variables, which provide justification for causal claims. However, suboptimal decisions in the design and analysis of experiments can produce invalid causal conclusions⁹³, and even well-designed experiments may face challenges⁹⁴, such as non-compliance or non-random dropout. Moreover, in ecology, experiments may be prohibitively expensive at the scales needed to detect causal effects, or they may distort natural ecological conditions⁸⁴, making them impractical or unrepresentative.

When experiments are infeasible, impractical, or unethical, observational designs for measured and unmeasured confounders are available. Advances in causal approaches for observational studies provide statistical techniques to satisfy causal assumptions without experimental manipulation^{12,22,25,77,95}. Observational designs for measured confounders rely on measuring all confounding variables^{9,74,75}. When measuring or even knowing all relevant confounders is not feasible, we can use observational designs for unmeasured confounders. These designs relax the causal sufficiency assumption of no unmeasured confounders by replacing it with assumptions about the structure of unmeasured confounders, typically informed by pre-existing knowledge. These designs then use statistical techniques to represent the influence of confounders based on their assumed structure^{96,97}, without needing to directly measure the confounders.

Experimental and observational designs can be implemented using either cross-sectional or longitudinal data. However, strong assumptions about temporal ordering (cause must precede its outcome) and stable effects over time are required to quantify causal effects using cross-sectional data.

Once data are collected, we can quantify the causal effect of interest using a range of estimation methods ("Collect Data and Apply Estimation Methods" and "Obtain Results" in Fig. 3). Many estimation methods are available to implement a chosen study design, each providing a different statistical approach for estimating the causal effect of interest^{98,99}.

After estimating a causal effect, we must then interrogate the plausibility of the causal assumptions underlying the study design and explore the implications of violations to these assumptions ("Interpret Results" in Fig. 3). One common approach for assessing the implications of violations is to perform causal sensitivity analyses, which quantify how an estimated effect would change in the presence of unaddressed confounding. Many sensitivity analysis techniques are available for a variety of causal inference methods^{100–104}, including SEM¹⁰⁵. An alternative approach to interrogating the plausibility of causal assumptions involves detecting under-adjustment of confounding variables by drawing on pre-existing knowledge to formulate tests of known effects^{11,106,107} (e.g., falsification or placebo tests). We must also consider how other forms of bias^{108,109}, such as selection bias^{110,111} and measurement bias^{112–114}, may influence the estimated effects and the robustness of our conclusions.

Algorithms for causal discovery

Algorithms for causal discovery fall into four categories: DC-based algorithms and three types of SCM-based algorithms, which are called constraint-based, score-based, and functional model-based algorithms. Representative algorithms from these four categories are listed in Table 2 (references, applications, and available software implementations are provided in Table S7). DC-based methods are suited for dynamic systems and assess causal relationships based on predictability and information flow over time. Constraint-based methods use conditional independence tests to eliminate implausible causal

Table 1 | Descriptions of study designs for causal inference

Category	Representative approaches ^a	Description
Experimental designs	Randomized controlled trials	Randomly assign units, or clusters of units, to treatment or control groups, which can address all confounders across groups (“all” = both measurable and unmeasurable confounders).
	Factorial designs	Randomly assign units to combinations of treatments, which can address all confounders across all treatment combinations (interactions).
	Crossover trials	Assign units to treatment and control conditions in a random sequence, which can address all confounders by allowing each unit to serve as its own control.
Observational Designs: Controlling measured confounders	Regression adjustment	Reweight observations using regression models, which can address measured confounders across groups exposed to different values of a cause (“cause” = a causal variable).
	Multi-level modeling with mixed effects	Reweight observations using regression models, which can address measured confounders (fixed effects) across groups exposed to different values of a cause.
	Structural equation modeling	Reweight observations using systems of regression models, which can address measured confounders across groups exposed to different values of a cause. ^b
	Marginal structural modeling ^c	Reweight observations over time using regression models, which can address measured confounders across groups exposed to different values of a cause when those values vary over time and confounders may be affected by past values of the cause.
	Subgroup (stratified) analysis	Reweight observations by grouping units into subgroups made during study design (e.g., stratified sampling) or analysis (e.g., subgroup comparisons), which can address measured confounders across groups exposed to different values of a cause.
	Covariate and propensity score matching	Reweight observations by matching units on their probability of being exposed to a specific value of the cause (propensity score matching) or on a metric of similarity in the values of confounders (e.g., Mahalanobis distance metric), which can address measured confounders across groups exposed to different values of a cause. Inverse propensity scores (i.e., Inverse Probability Weighting, IPW) can also be used as alternative weights in other approaches, like regression models.
Observational Designs: Controlling unmeasured confounders	Instrumental variables	Use a variable that affects the cause but has no direct effect on the outcome, which can address all unmeasured confounders for a subset of the units (units called compliers).
	Regression discontinuity design	Use a variable that, at specific values, creates a discontinuous change in the value of a cause but has no effect on the outcome, which can address all unmeasured confounders for a subset of the units (units near the discontinuity value).
	Front-door criterion	Use measured variables that comprise all intermediate variables on the causal path between a cause and an outcome, which can address all unmeasured confounders across groups exposed to different values of a cause.
	Before-after-control-impact ^c	Use within-unit, temporal variation in the cause within a subset of units, which can address unmeasured confounders that are constant across time or are varying at scales larger than the unit. The BACI approach is also known as Difference-in-Differences. Extensions exist, such as two-way fixed effects and matrix completion methods, which rely on less restrictive causal assumptions than BACI.
	Multi-level modeling with fixed effects ^c	Use within-unit, temporal variation in the cause within a subset of units, which can address unmeasured confounders that are constant across time or are varying at scales larger than the unit.
	Synthetic control methods ^c	Construct synthetic control groups from a weighted combination of unexposed units, which can address unmeasured confounders across groups exposed to different values of a cause.
	Interrupted time-series analysis ^d	Use a sudden change in a cause from a known source, which can address unmeasured confounders from pre-existing trends in the outcome.
	Principal stratification	Reweight observations by grouping units based on their potential values of a post-treatment variable (e.g., attrition), which addresses unmeasured confounders that operate through the post-treatment variable.

^aIn practice, multiple approaches can be combined to more credibly satisfy causal assumptions.

^bWith additional assumptions, SEMs can incorporate unobserved constructs (i.e., latent variables), which are inferred from measured variables.

^cRequires longitudinal data for which the value of the causal variable varies within and across units.

^dRequires longitudinal data for which the value of the causal variable varies within units.

relationships. Score-based methods evaluate possible graph configurations that represent causal interrelationships using a scoring criterion that captures how well the graph fits patterns of conditional independencies in the data. Functional model-based methods assume specific functional relationships between variables (e.g., linear or nonlinear equations with noise) and infer causal direction by identifying which graph configuration satisfies those assumptions.

Causal discovery algorithms have been developed to accommodate different data structures, with approaches often tailored to either longitudinal data or cross-sectional data. DC-based methods

require bivariate or multivariate time-series data (i.e., regularly spaced longitudinal data) to infer causal relationships through changes over time^{27,63}. In contrast, SCM-based algorithms can be applied to both cross-sectional and longitudinal data, but additional assumptions about temporal ordering (i.e., causes precede their outcomes) must be satisfied when using cross-sectional data^{28,76,115}. As with causal inference, pre-existing knowledge can enhance results from SCM-based discovery methods by explicitly specifying certain relationships that should or should not be included in the causal diagram.

Table 2 | Descriptions of algorithms for causal discovery

Category	Representative algorithms	Description
Constraint-based methods	PC (Peter and Clark)	Uses repeated conditional independence tests to infer causal relationships from observed independencies in data, producing a set of causal graphs that represent possible causal relationships consistent with the data.
	FCI (Fast Causal Inference)	Extends the PC algorithm to detect possible unmeasured confounders, producing a causal graph that reflects uncertainty about edges.
	PCMCI (Peter and Clark Momentary Conditional Independence)	A time-series adaptation of PC that improves detection of causal effects in auto-correlated data by iteratively testing for conditional independencies among variables and their lags.
Score-based methods	GES (Greedy Equivalence Search)	Searches for the best causal graph by iteratively adding or removing edges based on a scoring criterion, such as the Bayesian Information Criterion (BIC), balancing data fit and simplicity.
	GIES (Greedy Interventional Equivalence Search)	An extension of GES that incorporates interventional data or assumptions to distinguish between equivalent causal graphs.
	FGES (Fast Greedy Equivalence Search)	A variant of GES that uses a parallelized greedy approach to rapidly search for the optimal causal graph, making it suitable for high-dimensional datasets.
Functional model-based methods	LiNGAM (Linear Non-Gaussian Acyclic Model)	Identifies causal direction among variables by assuming linear relationships and non-Gaussian noise.
	ANM (Additive Noise Model)	Assumes the outcome variable is an unknown function of the causal variable plus independent additive noise, which enables identification of causal direction in both linear and nonlinear settings.
	IGCI (Information Geometric Causal Inference)	Determines causal direction by analyzing asymmetries in the joint distributions of cause-effect pairs, without inherently controlling for or detecting unmeasured confounders or indirect causal effects.
Dynamical systems causality (DC)-based methods	Granger Causality (GC)	Tests whether past values of one time-series can predict future values of another, assuming linear relationships in time-series data.
	Information-Theoretic (IT) Causality	A class of nonparametric and model-based methods that infer direct causal relationships by quantifying how knowledge of one variable reduces uncertainty about the future states of another variable. Includes Transfer Entropy (TE) approaches.
	Convergent Cross-Mapping (CCM)	Uses state-space reconstruction to infer causal relationships in nonlinear systems by testing whether past states of the causal variable can reliably predict current states of another variable.
	Partial Cross-Mapping (PCM)	An extension of CCM that adjusts for potential unmeasured confounders to isolate direct causal relationships more accurately.

BOX 3

Best practices for transparently communicating results from causal analyses in ecology

- **Clearly state and justify all the causal assumptions required by the study design or algorithm.** Studies that explicitly state and justify the assumptions underlying their causal claims allow subject matter experts to more effectively evaluate the credibility of these assumptions and use that evaluation to refine subsequent research.
- **Frankly discuss the most likely sources of violations in causal assumptions that could invalidate the conclusions.** Transparency about potential unmeasured confounding variables or other violations of causal assumptions should be the norm in causal research.
- **Report how detected or quantified causal relationships change under the most plausible potential violations of causal and statistical assumptions.** Perfectly satisfying causal assumptions is unlikely in any study, and thus an assessment of the robustness of conclusions to violations is an essential component of all high-quality studies (for example, see the “Select a study design or algorithm, apply estimation methods, obtain results, and interpret results” section).

Once candidate causal diagrams have been obtained (“Collect Data and Apply Estimation Methods” and “Obtain Results” in Fig. 3), we must assess whether the causal assumptions of the chosen discovery algorithm are plausible for the ecological system under study and explore the implications of violations to these assumptions (“Interpret Results” in Fig. 3). To assess the reliability of conclusions drawn from the causal discovery process and to evaluate the robustness of the inferred causal relationships, sensitivity analyses that explore the stability of results across different parameter settings should be undertaken¹¹⁶.

Challenges and opportunities

Making valid causal claims from ecological data requires moving beyond analyses that use prediction- and association-focused models, which typically fail to represent the true underlying causal structures of ecological systems^{67,117,118}. It instead requires satisfying or carefully relaxing the causal assumptions that allow observed statistical dependencies to be interpreted as evidence of causal relationships.

The cornerstone of high-quality causal research is to deliberately and critically assess whether the requisite causal assumptions can be

plausibly satisfied or relaxed given the available pre-existing knowledge, the selected causal task, and the chosen study design or algorithm. While this assessment may seem daunting, especially given the complexity of ecological systems, advances in causal methodologies have demonstrated how the strength of causal claims can be more transparently communicated (see Box 3).

As causal methods evolve, new advances help us relax or probe untestable assumptions in challenging real-world settings, which expands the relevance and applicability of causal methods to the complexities of ecological systems. Ecologists are uniquely positioned not only to benefit from these advances but also to contribute meaningfully to their development. Ecologists' experiences with experimental study designs, multiscale complex systems, and the integration of biotic and abiotic processes offer valuable insights into widespread challenges in causal research, such as spatial interactions, down-scaling, and unit-to-unit causation.

By connecting the causal assumptions, tasks, frameworks, and methods that play essential roles in causal research, our workflow (Fig. 3) provides a set of best practices for investigating causal questions in ecology. The workflow emphasizes the role of causal assumptions, which help us to formalize pre-existing knowledge, align the causal task with the research objective, and select a study design or algorithm that satisfies those assumptions and guides data collection and analysis. Thus, our workflow not only supports ecologists in conducting rigorous and transparent causal research, but it also facilitates cogent discussions about potential unresolved weaknesses in prior studies, which can motivate new studies. Through an iterative application of the workflow, we can enhance the accumulation and synthesis of ecological knowledge.

As causal approaches become more accessible and adaptable, ecologists have an opportunity to refine long-standing questions, generate new theory, and develop credible causal explanations of the natural world.

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Author contributions

H.E.C. led the paper. H.E.C., L.E.D. and P.J.F. co-organized, and P.J.F. funded, the workshop in which J.E.K.B., H.E.C., L.E.D., J.R.F., P.J.F., M.-J.F., C.G., J.R., B.S., I.S., K.J.S., G.S., and B.vH. contributed to establishing the goals and emphases of the paper. H.E.C., L.E.D., and P.J.F. initiated the paper concept and framing. H.E.C. and P.J.F. wrote the main text. J.E.K.B., L.E.D., J.R.F., M.-J.F., J.R., B.S., I.S., K.J.S., G.S., and B.vH. suggested edits to the drafts of the paper. H.E.C. conceived and wrote the Supplementary Information.

Competing interests

The authors declare no competing interests.

Additional information

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